

EDS 223 Assignment 1

Map making practice

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Air Quality and Cancer risk in Oahu, Hawaii via EJScreen Tool

The two maps were created to focus on the environmental injustice for those living on the island of Oahu in Hawaii that may face a higher risk of cancer from toxic chemicals in the air. The first map displays the percentage of high risk of cancer, as listed by the EPA's Toxics Release Inventory (TRI) Program, which is mostly concentrated at the bottom of Oahu. In the EJScreen technical documentaion (2023), the term "cancer" is defined as the "lifetime cancer risk from inhalation of toxics". The second map focuses on the toxicity concentration from the RSEI model that was found in the air. The raw values are lard ranging from 0 to 4,512,172.1, with Oahu having the max at 4,512. Thus, map 1 shows the percentile of high risk areas in danger of cancer from toxic chemicals, which was calculated by persons per million over a 70-year lifetime", with the max in Oahu being 50 per million. In simpler terms, there would be an estimated 50 additional cancer cases per million over a 70-year lifetime according to EPA. For map 2, percentiles are used, " with a median at the 50th percentile, which corresponds to the raw value of 603.032"(EJScreen, 2023). In the case for Oahu, the highest percentile roughly 78th percentile or a raw value at 1,078.

The burden of those living near the areas with high risk of cancer may impact those who face poverty, which could be further mapped to show those living in the vicinity with the highest percetange faciing high risk of cancer. Lastly, it's important to note that map 1 and map 2 have a strong correlation with the location of two major infrastructures on the island that pollute the island. These two are the Daniel K. Inouye International Airport and Par Hawaii Refining,an energy plant, that had a recent settlement in 2022 due to Clean Air Violations with the EPA. These two areas are also major tourist attractions, with a strip of hotels and multiple shopping areas increasing the amount of anthropogenic pollution emitted on the island.

Load packages

```
library(tidyverse) # For R
library(sf) # Read vector
library(here) # Locate files
library(stars)
library(tmap) # For static and interactive maps
library(janitor) # Clean data
```

Read EJScreen and Data

```
# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data",
  "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"),
  quiet = TRUE)

# filter to a state you are interested in
hawaii <- ejscreen |>
  dplyr::filter(ST_ABBREV == "HI")

# filter to a county you are interested in
honolulu <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("Honolulu County"))

# find the average values for all variables within counties
hawaii_counties <- aggregate(hawaii,
  by = list(hawaii$CNTY_NAME), FUN = mean)
```

Data Cleaning

```
# Convert all columns to lower case
hawaii <- hawaii |>
  clean_names()

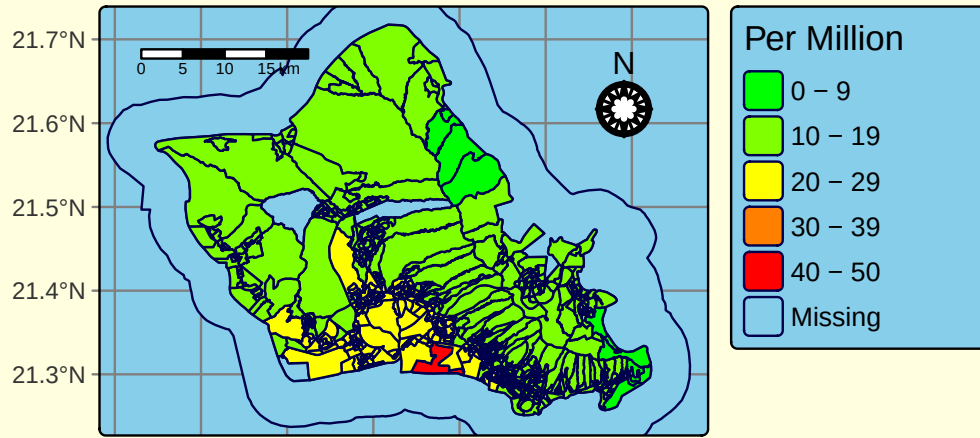
honolulu <- honolulu |>
  clean_names()
```

```
hawaii_counties <- hawaii_counties |>
  clean_names()
```

Map 1: Percentage of Lifetime Risk of Cancer from Toxic Air

```
# Select primary spatial object as county of choice
m1 <- tm_shape(honolulu,
bbox = c(-158.3042, 21.2367, -157.6251, 21.7293)) +
  tm_graticules() +
  tm_polygons(fill = "cancer", # Map percentile of air cancer
    fill.legend = tm_legend(title = "Per Million", # Create legend
    bg.color = "skyblue",
    width = 8),
    fill.scale = tm_scale_intervals(
      values = c("green", "yellow", "red"),
      value.na = "skyblue")) +
  tm_compass( # Customize compass
    type = "rose",
    size = 1.5,
    position = c(
      "right", "top")) +
  tm_scalebar(position = c("left", "top"),
    text.size = .4) +
  tm_title(
    text = "Lifetime Cancer Risk from
    Toxic Air in Oahu Per Million (EJScreen, 2019)",
    size = 1) + # Add title to map
  tm_layout(scale = 1.2,
    asp = 0, # Customize backgrounds
    style= "beaver",
    bg.color = "skyblue",
    outer.bg.color = "lightyellow")
m1
```

Lifetime Cancer Risk from
Toxic Air in Oahu Per Million (EJScreen, 2019)



Map 2: Toxic Air Chemicals

```
m2 <- tm_shape(honolulu,
bbox = c(-158.3042, 21.2367, -157.6251, 21.7293)) +
  tm_graticules() +
  tm_polygons(fill = "p_rsei_air",
    fill.legend = tm_legend(title = "Percentile",
      bg.color = "skyblue",
      width = 8),
    fill.scale = tm_scale_intervals(
      values = c("green", "yellow", "red"),
      value.na = "skyblue")) +
  tm_compass(
    type = "rose",
    size = 1.5,
    position = c("right", "top")) +
  tm_scalebar(position = c("left", "top"),
    text.size = .4) +
  tm_title(text = "Percentile of Toxic Chemicals in Oahu (EJScreen 2019)",
    size = 1) +
  tm_layout(scale= 1.2,
```

```

asp = 0,
style = "beaver",
bg.color = "skyblue",
outer.bg.color = "lightyellow")

```

m2

